



The typical wireless keyboard and mouse, which are the norm today

keep popping up, there's hope for those of us who want to be able to sit back instead of hunching over our keyboards and reaching out for that oddly-placed mouse!

From all the news and prototypes we've come across, at least a few revolve around changing the way we interface with PCs. The research points in four basic directions: better touch-based devices (as in new designs for glorified keyboards and mice), speech control, eye control, and even thought control. Basically, we're looking for computers to develop the senses we have—touch (they already feel us typing and clicking), hearing (speech control), vision (eye control), and extra-sensory perception in terms of thought control. Smell and taste we might not want them to develop—the last thing you need is your PC telling you that you smell, or chomping down on your finger if it gets hungry!

Listen To Me

Speech recognition has been around for quite some time now, but isn't really going anywhere far. We've reached a point where computers can understand us when we talk—at least as far as understanding commands goes—but that's provided you talk in an expressionless monotone, and are prepared to spend hours training the software. And don't forget that you always have to keep shushing people because the mic cannot tolerate background noise.

The shortcomings of speech-to-text are well documented, and we don't need to focus on it here. All we can hope for is adaptive or intelligent software that will be able to understand what we say, regardless of whether we have a cold, are breathless, or are talking to our PDAs in noisy railway stations!

Watch Me

The more interesting alternative input devices being researched are the visual-aided ones. Eye-control or gesture control devices use a camera to interface with customised software, and translate your motion into mouse control.

Interestingly, this was first thought of to help the physically disabled interface with PCs. However, as technology progresses, and as we all become a little disabled in our own way (basically, lazy!), such input techniques might come as a boon to us all.

Most devices actually track deviations in your eyes or the entire head. Some of these setups let you left-click with a wink of your left eye and right-click with your right eye; double-clicking is taken care of by rapid blinking. The

problem with such devices is that you're generally confined to your chair and cannot move your head more than a few inches to the left or right. This is because the camera needs to focus on your face or head. Now, for the physically handicapped, such restrictive devices make perfect sense, but not for the majority.

Take TrackIR 4 Pro (www.eyecontrol.com) for instance, a head-control system by NaturalPoint Inc. This system consists of a little headpiece with three reflective surfaces, and which can be attached to almost any headwear, such as a baseball cap. It has a sensor system that, with the help of software, tracks the way your head moves and translates it to movements in games. So instead of moving your mouse to look left, you can just look left!



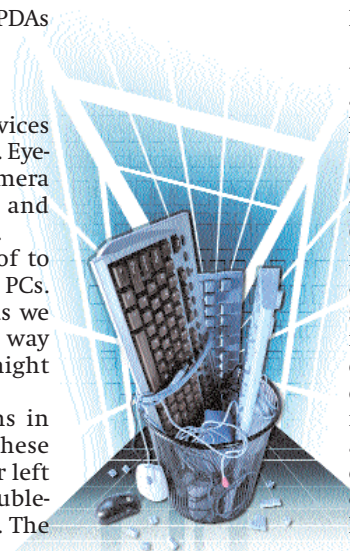
TrackIR 4 Pro translates head movements into mouse movements on your screen

"But doesn't that mean looking away from the monitor?" is the question everyone asks. The answer is simple: sensitivity! You can set the software to be less or more sensitive to your movements, so moving your head left by a degree could change your view in the game by, say, 10 degrees.

What's more, TrackIR 4 Pro looks for movements along all three axes, and also tracks movements in the form of pitching, yawing, and rolling. This is ideal for, say, a game like Microsoft's *Flight Simulator*.

Reviewers from around the world claim that this form of control is very easy to get used to, and that it shouldn't take you more than an hour to be in complete control!

Now if the TrackIR system is built specifically for gaming, there are other products built for regular computer use. An example is Quick Glance from www.iriscom.org, which is specifically targeted at people unable to use a mouse and keyboard. It's a completely custom-built solution that comes with training, so not too much detail can be provided here. However, it does use eye-tracking to move a pointer, which can click on a particular area either by just leaving it over an icon for a short while (by staring at the icon), or by blinking slowly. As you can expect, the system comprises software and a camera that can be mounted on your desktop or laptop monitor.



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